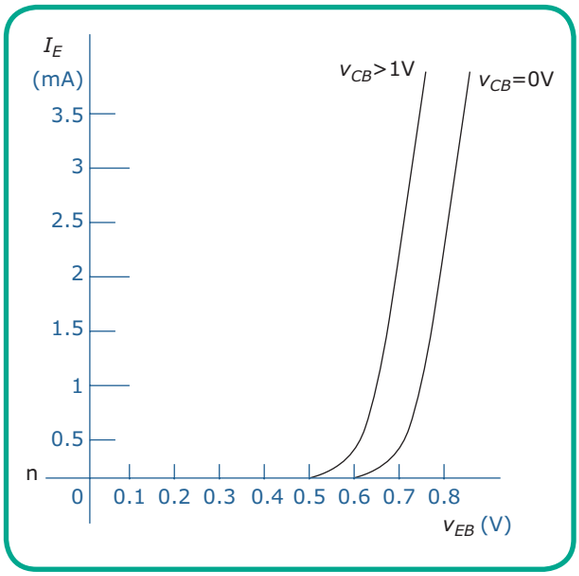


Figure 5.14 Input characteristics

Output Characteristics

It is defined as the characteristic curve drawn between output voltage to output current whereas input current is constant.

To determine output characteristics, the emitter current I_E is kept constant at zero and collector current I_c is increased from zero by increasing V_{CB}. This is repeated for higher fixed values of I_E.

A curve is drawn between the output voltage (V_{CB}) to output current (I_C) at constant input current (I_E) as shown in the figure 5.15.

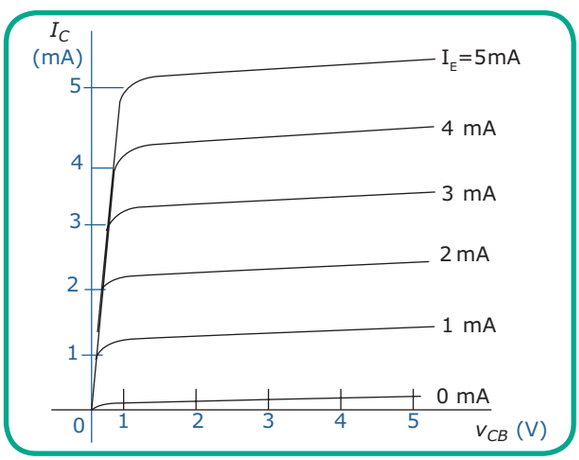


Figure 5.15 Output characteristics

Current Amplification Factor (α).

It is the ratio of output current to input current. In common base connection, the input current is the emitter current I_E and output current is the collector current I_C.

The ratio of change in collector current (ΔI_C) to the change in emitter current (ΔI_E)at constant collector base voltage V_{CB} is known as current amplification factor i.e.

$$\propto = \frac{\Delta I_C}{\Delta I_E} \text{ at constant } V_{CB}$$

5.9.2. Common Emitter Configuration

Common emitter configuration is shown in the figure 5.16. Here emitter is grounded and it is used as the common terminal for both input and output. It is also called as grounded emitter configuration. Base is used as an input terminal whereas collector is the output terminal.

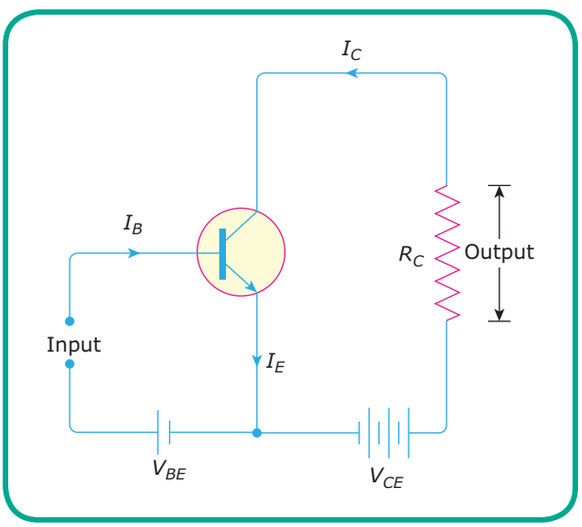


Figure 5.16 Common Emitter configuration

Input Characteristics

It is defined as the characteristic curve drawn between input voltage to input current whereas output voltage is constant.

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